Peer review and response

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Address the concern, preserve original records and distinguish proposed analysis from completed evidence.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Writing results and limitations. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Figures that preserve **evidence**. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

Writing results and limitations

> **My remembered idea, example and correction**

Figures that preserve evidence

> **My remembered idea, example and correction**

Inspecting failures and anomalies

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Peer review tests evidence, methods and clarity. For each comment record concern, response, change location, evidence and status. Main reviewer requests a fitted-line baseline, while the author only calls a neural network advanced. That supplies no comparison. Add an explicitly planned fair comparison and label it new analysis, or narrow the claim to existing evidence. Proposed tests are not completed results.

  Separate picture has four views of P1 and four of P2: eight photos from two plants. Eight independent-plant observations are unsupported. Revisit grouping, data-role allocation and uncertainty assumptions, not merely wording. Different IDs do not establish sampling independence, and no labels/split assignments are supplied for a revised rate or actual leakage diagnosis. Correct grouping cannot create more evidence.

  Missing failure counts need real records/denominators or an honest statement of absence, not deletion of limitations. This is a fictional paper response exercise, not a message to a reviewer.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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  Missing failure counts need real records/denominators or an honest statement of absence, not deletion of limitations. This is a fictional paper response exercise, not a message to a reviewer.

- Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison. Show the intermediate steps and give the conclusion with its scope.

Main response step	Evidence-grounded alternative
Identify concern	Does performance differ from a fitted-line baseline under aligned conditions?
Reject unsupported defense	Advanced architecture is not comparative evidence
If no comparison exists	Narrow the claim to the actual recorded comparison
If new comparison is proposed	Specify aligned cases/metric/eligibility and label new analysis
Change location	Identify the actual manuscript section/artifact when revised; do not invent it
Status	Proposed, completed or blocked, with corresponding evidence

Main response step	Evidence-grounded alternative
Preserve history	Original claim, response and later analysis remain distinguishable

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Peer review tests evidence, methods and clarity. For each comment record concern, response, change location, evidence and status. Main reviewer requests a fitted-line baseline, while the author only calls a neural network advanced. That supplies no comparison. Add an explicitly planned fair comparison and label it new analysis, or narrow the claim to existing evidence. Proposed tests are not completed results.

  Separate picture has four views of P1 and four of P2: eight photos from two plants. Eight independent-plant observations are unsupported. Revisit grouping, data-role allocation and uncertainty assumptions, not merely wording. Different IDs do not establish sampling independence, and no labels/split assignments are supplied for a revised rate or actual leakage diagnosis. Correct grouping cannot create more evidence.

  Missing failure counts need real records/denominators or an honest statement of absence, not deletion of limitations. This is a fictional paper response exercise, not a message to a reviewer.

- A report omits this qualification: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Peer review tests evidence, methods and clarity. For each comment record concern, response, change location, evidence and status. Main reviewer requests a fitted-line baseline, while the author only calls a neural network advanced. That supplies no comparison. Add an explicitly planned fair comparison and label it new analysis, or narrow the claim to existing evidence. Proposed tests are not completed results.

  Separate picture has four views of P1 and four of P2: eight photos from two plants. Eight independent-plant observations are unsupported. Revisit grouping, data-role allocation and uncertainty assumptions, not merely wording. Different IDs do not establish sampling independence, and no labels/split assignments are supplied for a revised rate or actual leakage diagnosis. Correct grouping cannot create more evidence.

  Missing failure counts need real records/denominators or an honest statement of absence, not deletion of limitations. This is a fictional paper response exercise, not a message to a reviewer.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Reviewer notes missing failure counts; author deletes the limitation paragraph. What is an adequate response?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled peer review and response **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of peer review and response.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Peer review and response. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Peer review and response. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Reviewer notes missing failure counts; author deletes the limitation paragraph. What is an adequate response?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

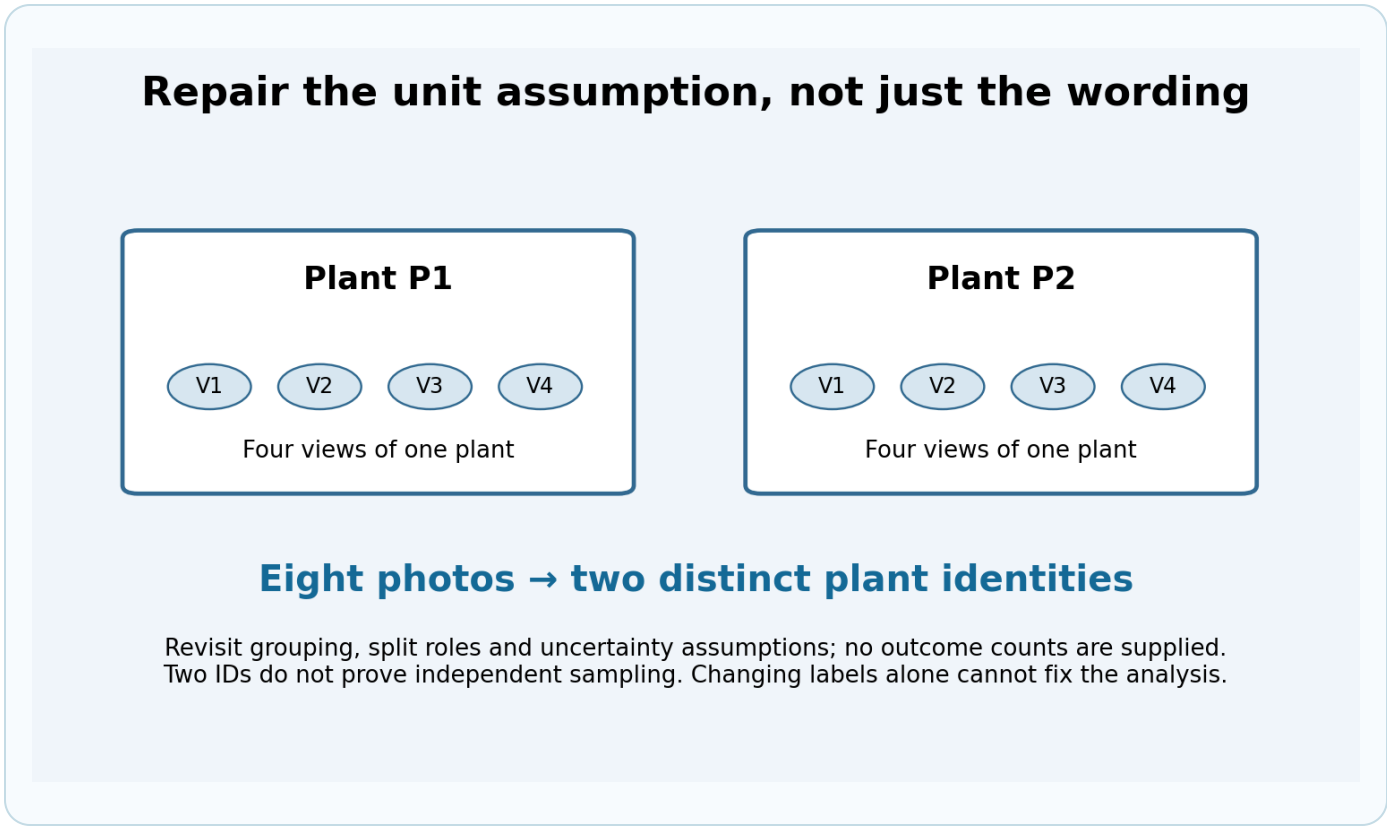
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Respond to a reviewer with **evidence**. A reviewer asks why four photos of one plant were treated as four independent plants.
- The original table contains four views of P1 and four views of P2: eight photos from two plants.
- Count photos and underlying plants in the reviewer's concern. What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/32

Peer review and response

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Identify the concern

The reviewer questions the unit of analysis.

2. Check the records

Plant identifiers confirm two plants, not eight independent plants.

3. Revise transparently

Correct the description and revise any analysis that assumed independence.

CHANGE ONE THING

The team only changes "photos" to "plants" without revisiting analysis.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Respond to a reviewer with **evidence**. A reviewer asks why four photos of one plant were treated as four independent plants.
- Which labelled step matches this detail? Correct the description and revise any analysis that assumed independence.
- 1. Revise transparently
- 2. Check the records
- 3. Identify the concern
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Revise transparently Correct the description and revise any analysis that assumed independence.

Peer review should connect a criticism to the actual records and a concrete correction or reasoned response. Polite language alone cannot repair a methodological error.

Changed-case question and optional hint

- Changed-case practice: The team only changes "photos" to "plants" without revisiting analysis. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The substantive independence issue remains unresolved.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The substantive independence issue remains unresolved. Renaming records does not change their shared source objects.

Repair and retain the record

Review Writing results and limitations (ai-7/31). Goal: Explain the **mechanism** and terminology of writing results and limitations. Short practice: Compute the supported-answer rates for 9/12 and 10/12.

What actual outcome was measured, and why is a claim that every student becomes smarter outside this **evidence**?. Return to this week and explain what you changed.

Review Figures that preserve **evidence** (ai-7/29). Goal: Explain the **mechanism** and terminology of figures that preserve **evidence**. Short practice: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?. Return to this week and explain what you changed.

Review Inspecting failures and anomalies (ai-7/26). Goal: Explain the **mechanism** and terminology of inspecting failures and anomalies. Short practice: Calculate mean and median for 2,2,2,10.

What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

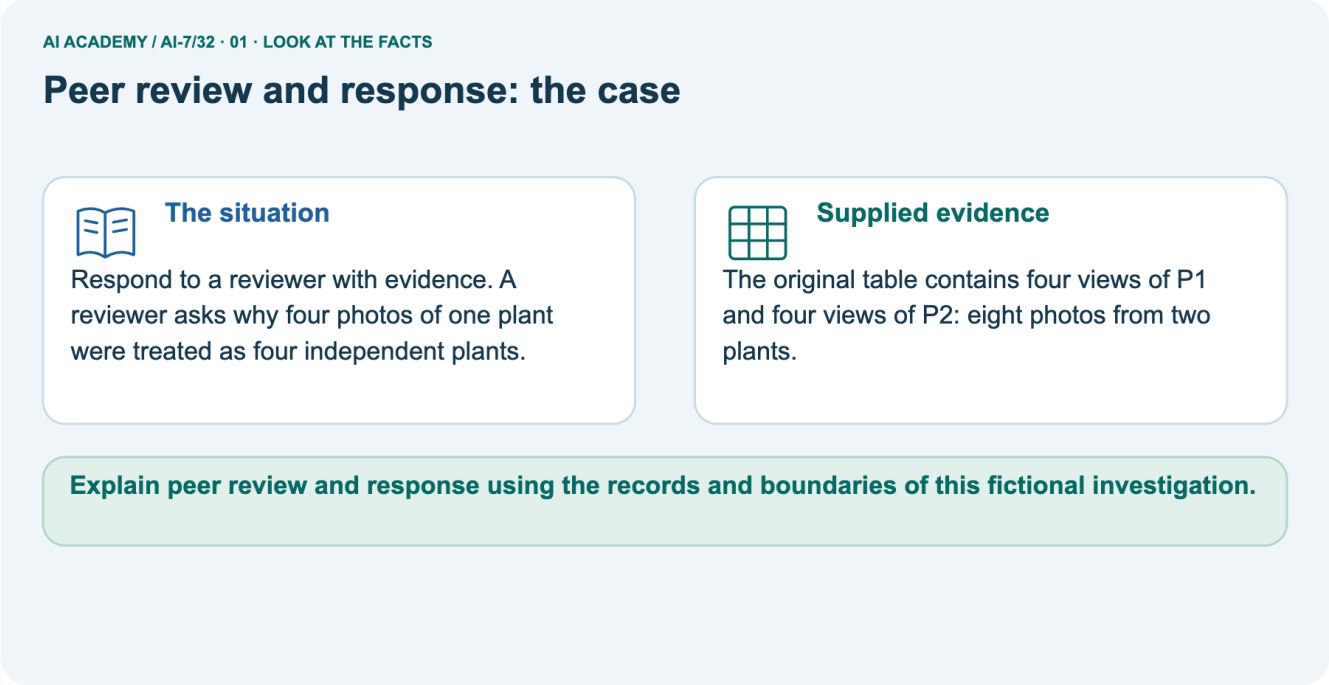
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Peer review and response: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

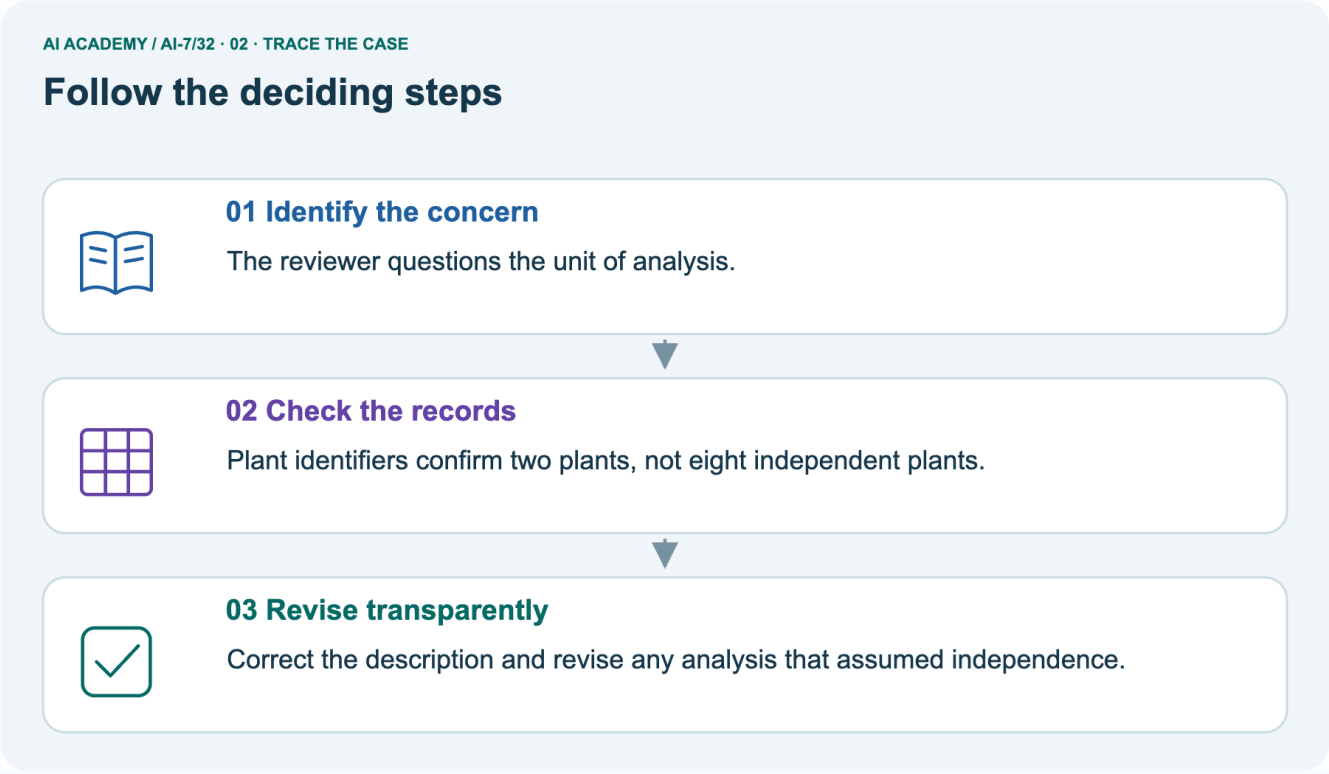
Respond to a reviewer with **evidence**. A reviewer asks why four photos of one plant were treated as four independent plants.

Supplied facts: The original table contains four views of P1 and four views of P2: eight photos from two plants.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



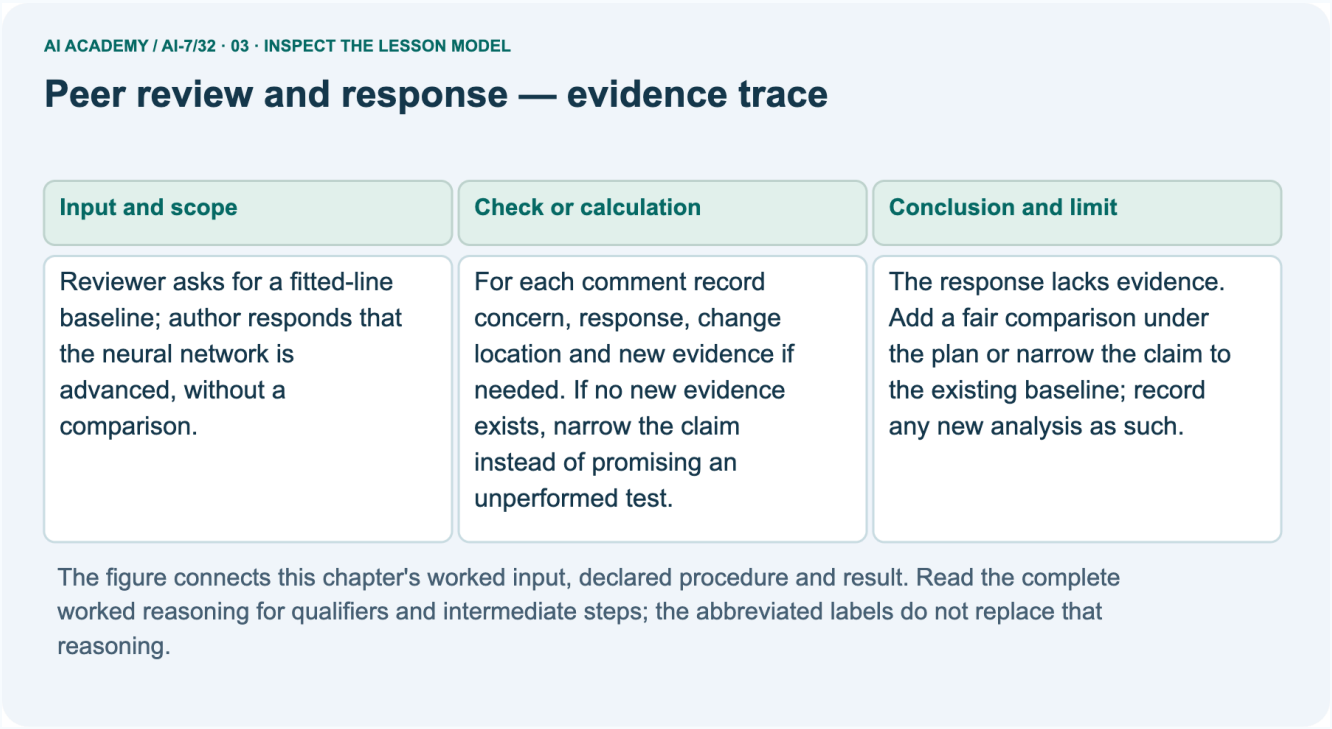
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the concern: The reviewer questions the unit of analysis. → Check the records: Plant identifiers confirm two plants, not eight independent plants. → Revise transparently: Correct the description and revise any analysis that assumed independence.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Peer review and response — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Peer review and response — **evidence** trace.

Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison. | For each comment record concern, response, change location and new **evidence** if needed.

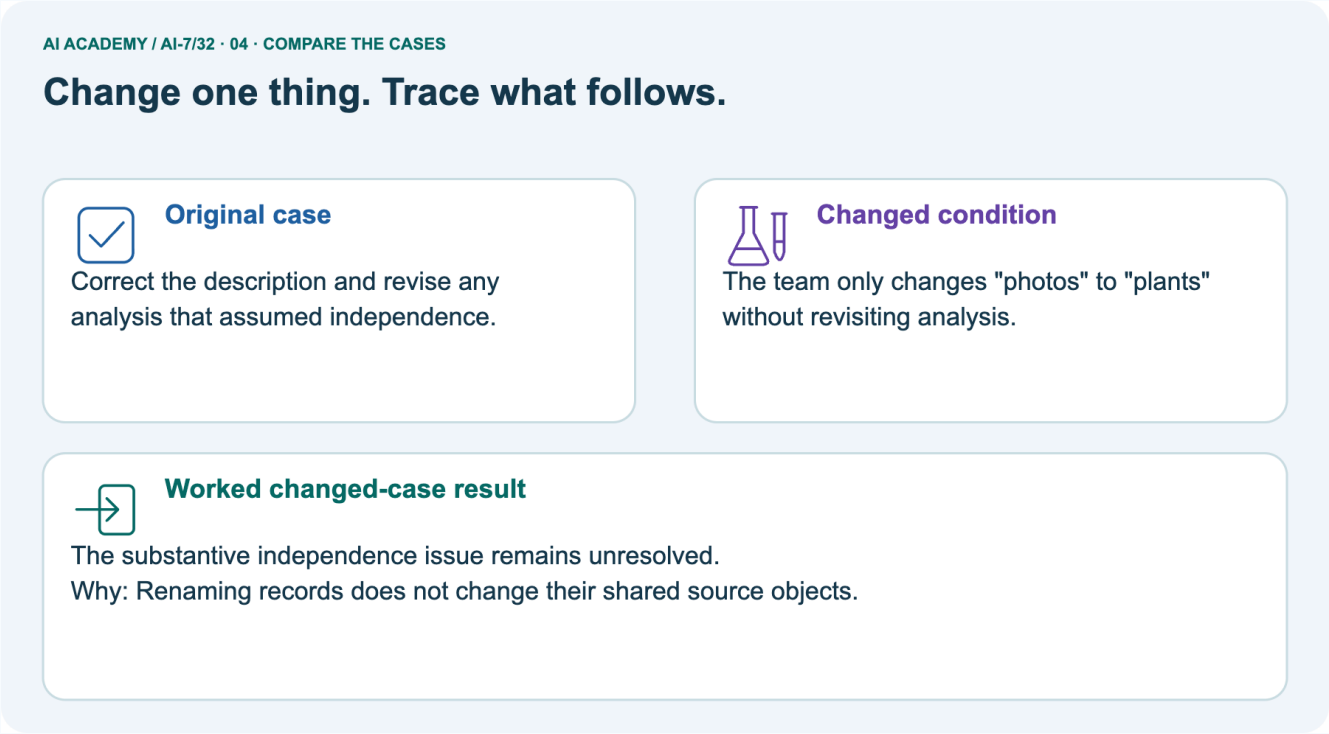
If no new **evidence** exists, narrow the claim instead of promising an unperformed test. | The response lacks **evidence**.

Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

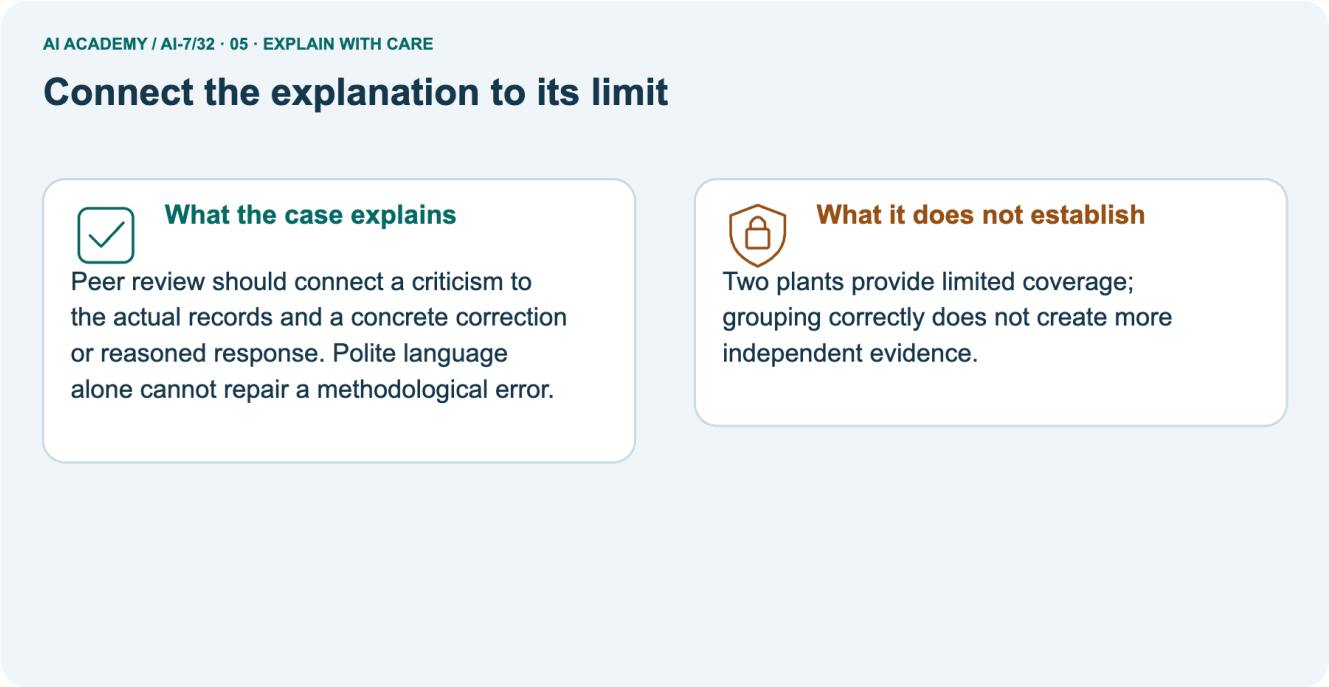
Original case: Correct the description and revise any analysis that assumed independence.
Changed condition: The team only changes "photos" to "plants" without revisiting analysis.

Expected result: The substantive independence issue remains unresolved. Reason: Renaming records does not change their shared source objects.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

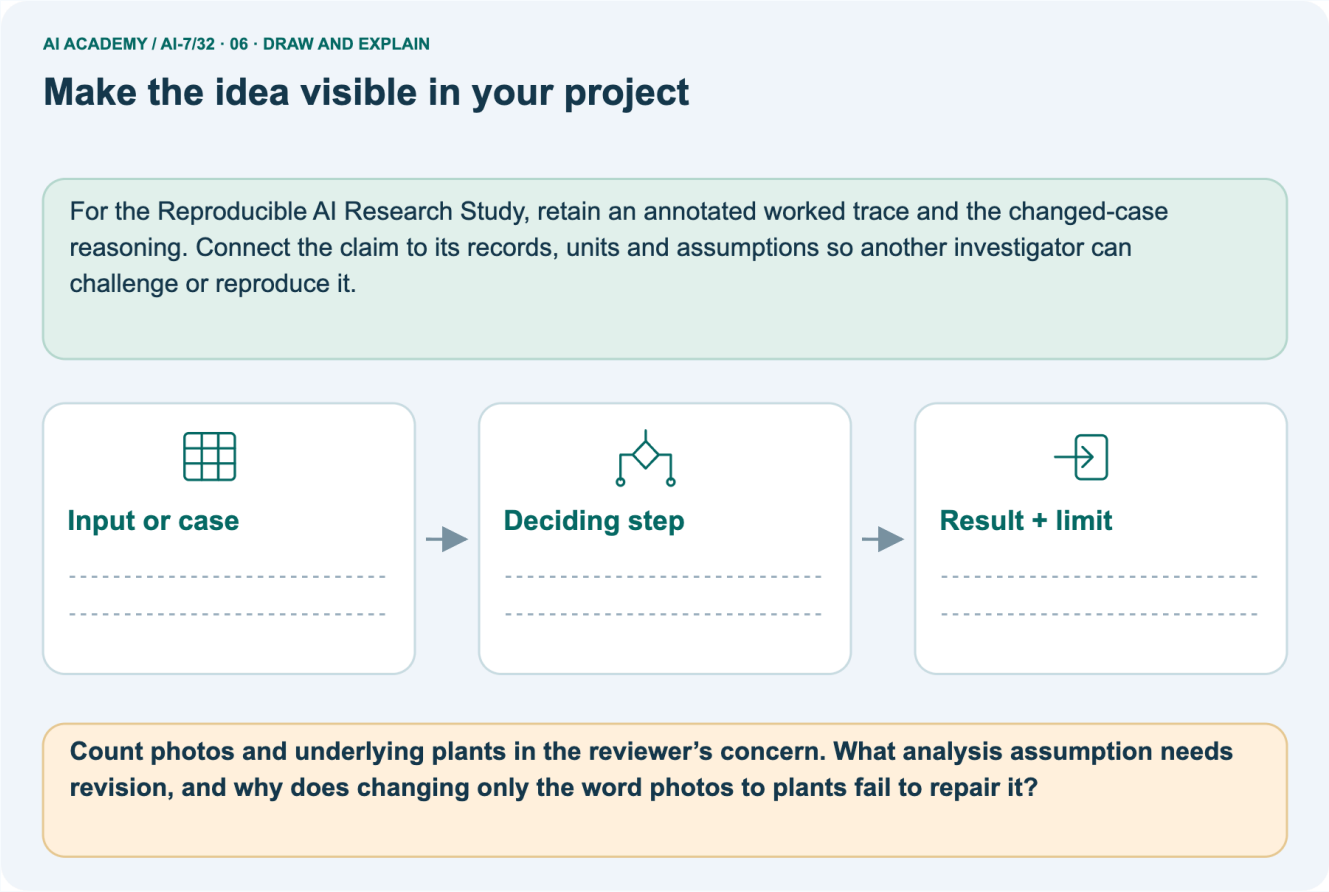
Peer review should connect a criticism to the actual records and a concrete correction or reasoned response. Polite language alone cannot repair a methodological error.

Limit: Two plants provide limited coverage; grouping correctly does not create more independent **evidence**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Count photos and underlying plants in the reviewer's concern.

What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Peer review and response: the case

02. Follow the deciding steps

03. Peer review and response — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.